Exercise – Understanding the t distribution using simulation

1. Aim of the exercise

The aim of this exercise is to understand the t statistic using simulation.

2. Distribution of the t statistic

In the exercise dedicated to the derivation of the t statistic, we have shown that the stochastic representation of the Student's t-distribution with $n-K$ degrees of freedom is given by

$$t_k \sim \frac{N(0,1)}{\sqrt{\chi^2[n-K]/(n-K)}} = t[n-K]. \tag{1}$$

It shows how a t-distributed random variable can be constructed from a standard Normal random variable and an independent chi-squared random variable.

We have also shown that the probability density function of the Student's t-distribution is given by

$$f(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \tag{2}$$

where ν denotes the number of degrees of freedom. In our case, $\nu = n-K$. The gamma function is expressed using Euler's integral:

$$\Gamma(x) = \int_0^\infty z^{x-1}\cdot e^{-z}\mathrm{d}z, \qquad x>0.$$

3. Define parameter values

Before simulating and analyzing the Student's t-distribution, we define the parameter values that will be used throughout the exercise. Specifically, we specify three degrees-of-freedom values, which will later allow us to examine how the shape of the distribution changes as the degrees of freedom increase. We also define a common range of x-values that will be used in all figures to ensure consistent scaling across histograms and density plots. Finally, we create histogram bin edges and a grid of evaluation points for plotting the theoretical probability density functions.

```
%% 2. Define parameter values

% 2.1. Define degrees of freedom
nu5 = 5;
nu9 = 9;
nu30 = 30;

% 2.2. Define common x-axis range
xMin = -6;
xMax = 6;

% 2.3. Define histogram bin edges
binEdges = xMin:0.01:xMax;
```

```
21
22 % 2.4. Define PDF evaluation points
23 x = xMin:0.01:xMax;
```

4. Simulating the t distribution from its stochastic representation

We use the stochastic representation of the Student's t-distribution given in equation (1) to generate random draws from the distribution. For this purpose, we make use of the custom function `StochasticRepresentation`. The function first generates independent draws from a standard normal distribution and a chi-squared distribution. These draws are subsequently combined according to equation (1) to obtain t-distributed random variables. The resulting values are stored in the variable `draws`. Figure 1 displays a histogram of the simulated draws.

```
25 %% 3. Simulating the t distribution from its stochastic representation
26
27 % 3.1. Number of random draws
28 N = 1000;
29
30 % 3.3. Construct t-distributed random variables
31 draws = StochasticRepresentation(N,nu5); % In MATLAB, random('T',nu,N,1)
32
33 % 3.4. Plot histogram
34 figure('Position',[100 100 1000 1000]);
35 histogram(draws,...
36     'BinEdges',binEdges,...
37     'FaceColor',[0.700 0.700 0.700],...
38     'EdgeAlpha',0,...
39     'DisplayName','All simulated values');
40 xlim([xMin xMax]);
41 title(['Fig. 1. Histogram of simulated t-distributed draws', ...
42        ', N = ',num2str(N), ...
43        ', \nu = ',num2str(nu5)]);
44 xlabel('t');
45 ylabel('Frequency');
46 legend('show');
```

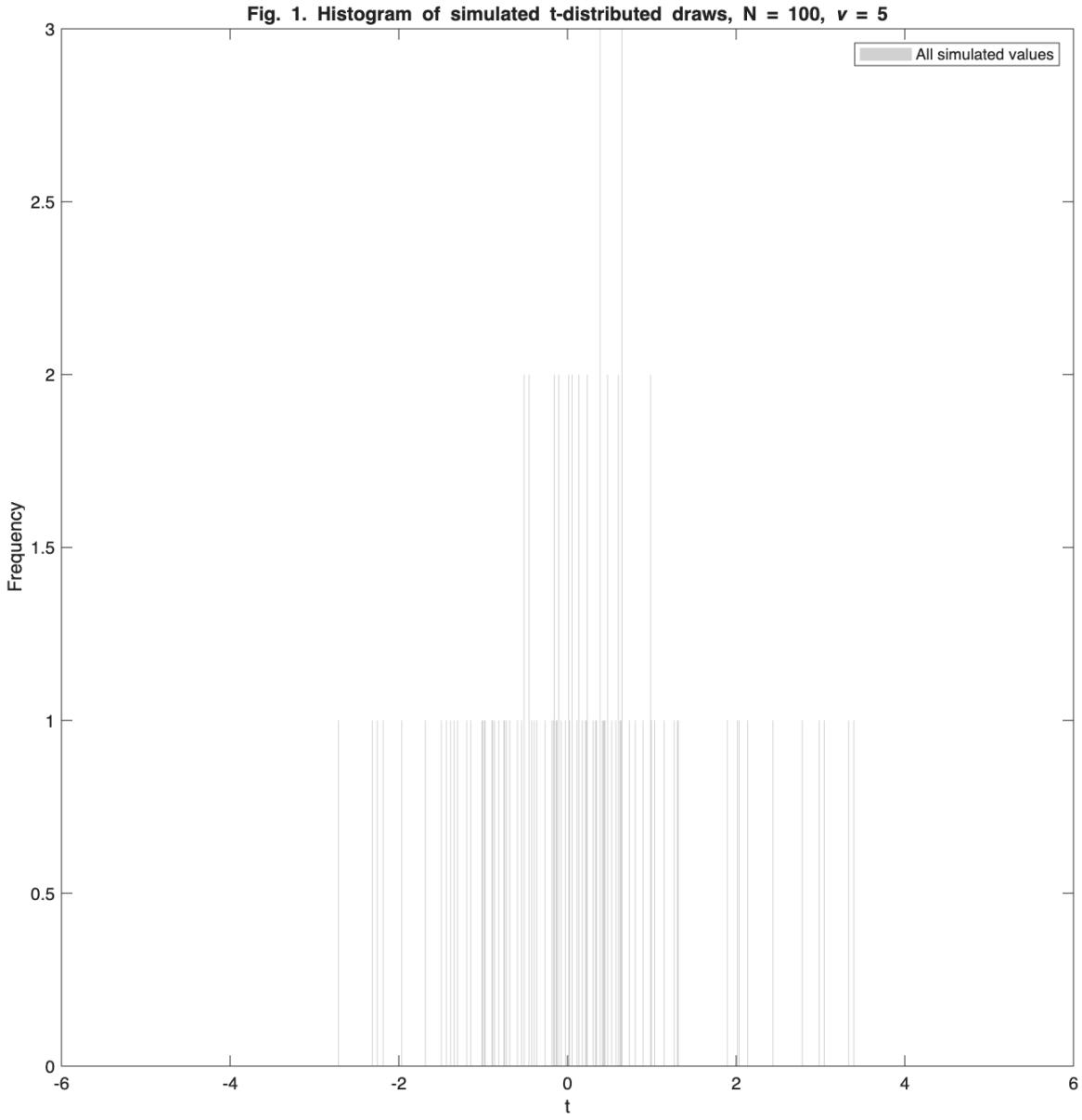


Figure 1: Frequency distribution of 100 draws

5. Increasing the number of simulated draws

Here, we increase the number of draws. Because the code is almost identical to that presented previously, only the modified lines are shown. The only change is that the number of draws is increased.

The resulting histogram is presented in Figure 2. As the sample size increases, the histogram becomes smoother and more closely resembles the theoretical shape of the Student's t-distribution. This illustrates that the empirical distribution of the simulated draws converges to the underlying population distribution as the number of draws increases. As more observations become available, random sampling variation has less influence on the shape of the histogram. Several characteristics of the t-distribution are therefore easier to observe. In particular, the distribution is centered around zero, symmetric, and bell-shaped.

The distribution is centered around zero because the Student's t-distribution is derived as the sampling distribution of a t-statistic under the null hypothesis. As explained in the exercise on the derivation of the t-statistic, the null hypothesis typically states that there is no effect or no difference, implying that the expected value of the statistic is zero. Consequently, positive and negative values occur with equal probability, giving rise to a symmetric distribution centered at zero.

The most important point to note is that the values shown in the histogram are not real-world observations. Rather, they are hypothetical draws generated from a statistical model under the assumptions that give rise to the Student's t-distribution. The histogram therefore represents what could happen across many repeated samples, not what did happen in a particular dataset. This distinction between hypothetical repeated samples and observed data is fundamental to the frequentist approach to statistical inference and will become important in later exercises.

```
49 %% 4. Increasing the number of simulated draws
50
51 % 4.1. Number of random draws
52 N = 1000000;
```

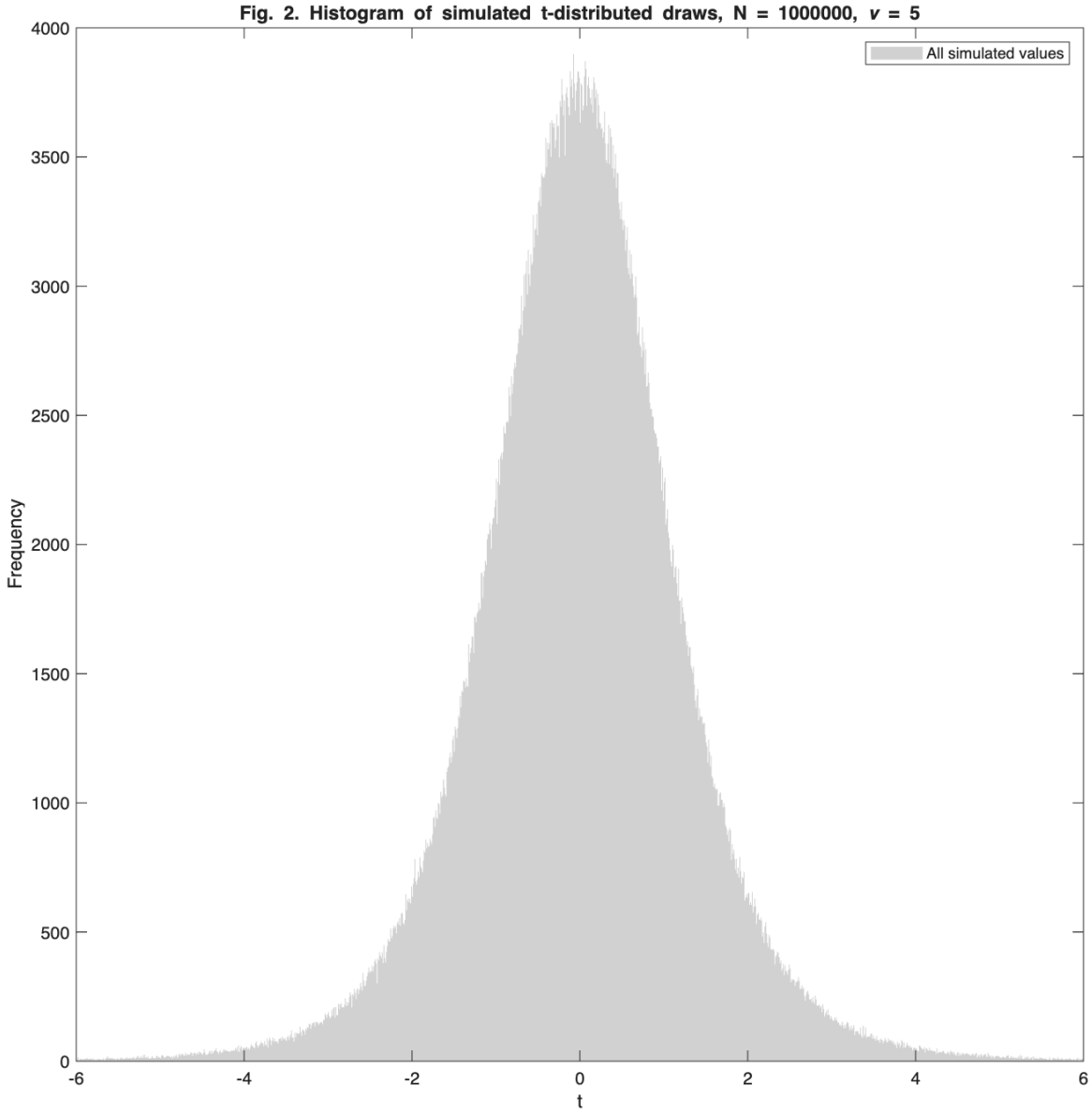


Figure 2: Frequency distribution of 1000000 draws

6. Evaluating the theoretical PDF of the t distribution

Instead of evaluating the probability density function manually at each value of `x`, we make use of the custom function `PDF`. This function implements the analytical expression of the Student's t-distribution and evaluates the gamma functions numerically using Euler's integral representation. We first define a grid of evaluation points, stored in `x`. The function `PDF` then computes the corresponding density values for a Student's t-distribution with 5 degrees of freedom. The resulting density values are stored in `y`.

Figure 3 shows the theoretical probability density function of the Student's t-distribution. Unlike the histograms presented earlier, which were based on simulated draws, this figure is obtained directly from the analytical formula of the distribution. The curve therefore represents the exact theoretical density implied by the model. As the number of simulated draws increases, the histogram approaches this theoretical density more closely.

```
%% 5. Evaluating the theoretical PDF of the t distribution

% 5.1. Evaluate the PDF of the t distribution
pdfNu5 = PDF(x,nu5); % In MATLAB, pdf('T',x,nu5)

% 5.2. Plot the PDF
figure('Position',[100 100 1000 1000]);
plot(x,pdfNu5,...
    'DisplayName','PDF');
xlim([xMin xMax]);
title(['Fig. 3. Theoretical PDF of the t distribution, \nu = ', ...
    num2str(nu5)]);
xlabel('t');
ylabel('Density');
legend('show');
```

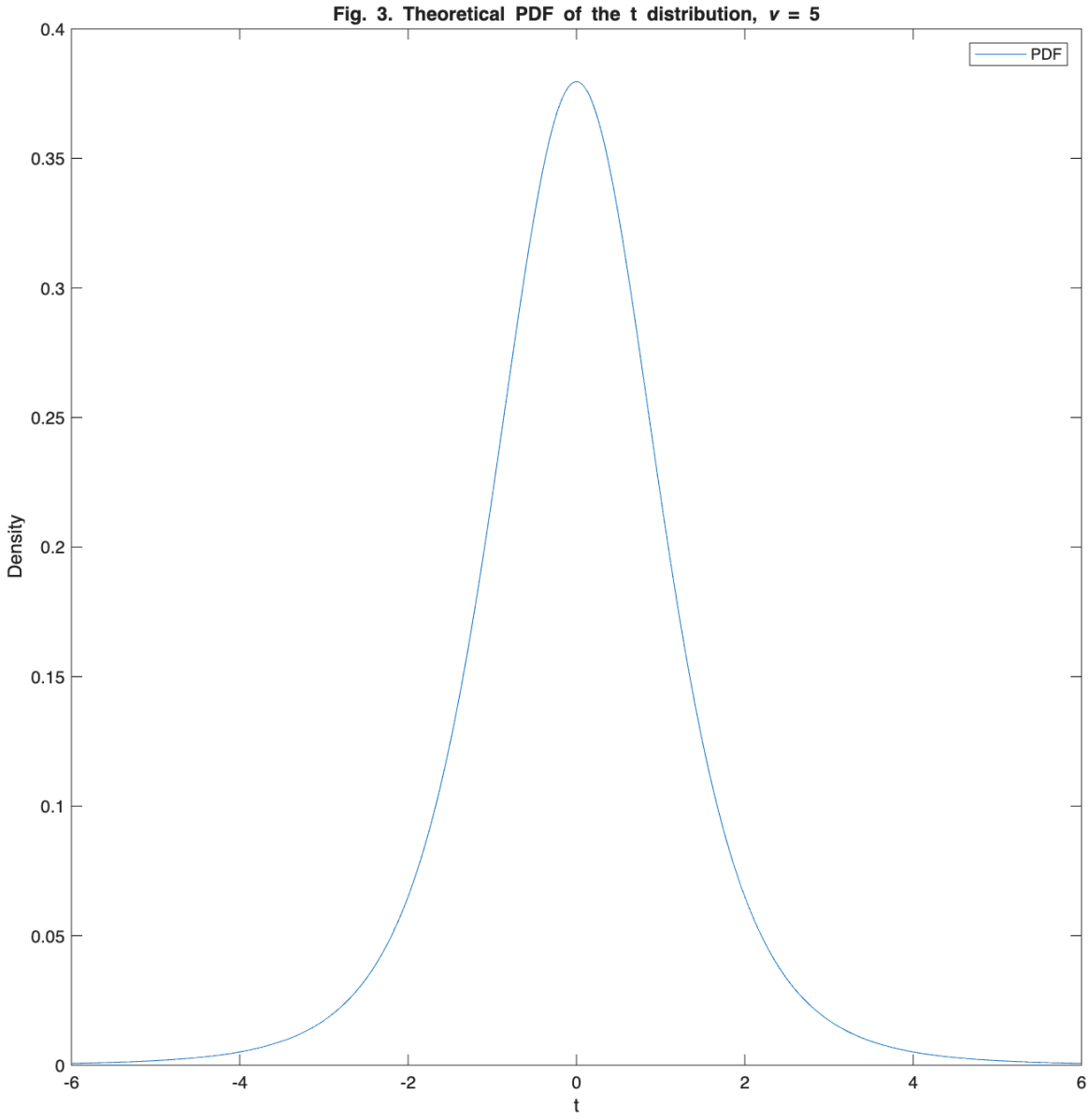


Figure 3: The limiting distribution

7. Effect of the degrees of freedom on the PDF

Next, we investigate how the shape of the Student's t-distribution changes as the degrees of freedom increase. In addition to the case of 5 degrees of freedom, we consider 9 and 30. For each value, we evaluate the theoretical probability density function using our custom `PDF` function. Since most of the code is identical to the previous example, only the lines that have been modified are shown. Specifically, we define additional degrees of freedom and evaluate the corresponding density functions.

The result is shown in Figure 4. As the degrees of freedom increase, the Student's t-distribution becomes increasingly similar to the standard normal distribution. The heavier tails gradually diminish, reducing the probability of extreme observations. In the limit as the degrees of freedom approach infinity, the Student's t-distribution converges to the standard normal distribution. This limiting behavior is already visible when the degrees of freedom equal 30, where the density is almost indistinguishable from the standard normal density.

```
90 %% 6. Effect of the degrees of freedom on the PDF
91
92 % 6.1. Evaluate the PDFs of the t distribution
93 pdfNu5 = PDF(x,nu5);
94 pdfNu9 = PDF(x,nu9);
95 pdfNu30 = PDF(x,nu30);
```

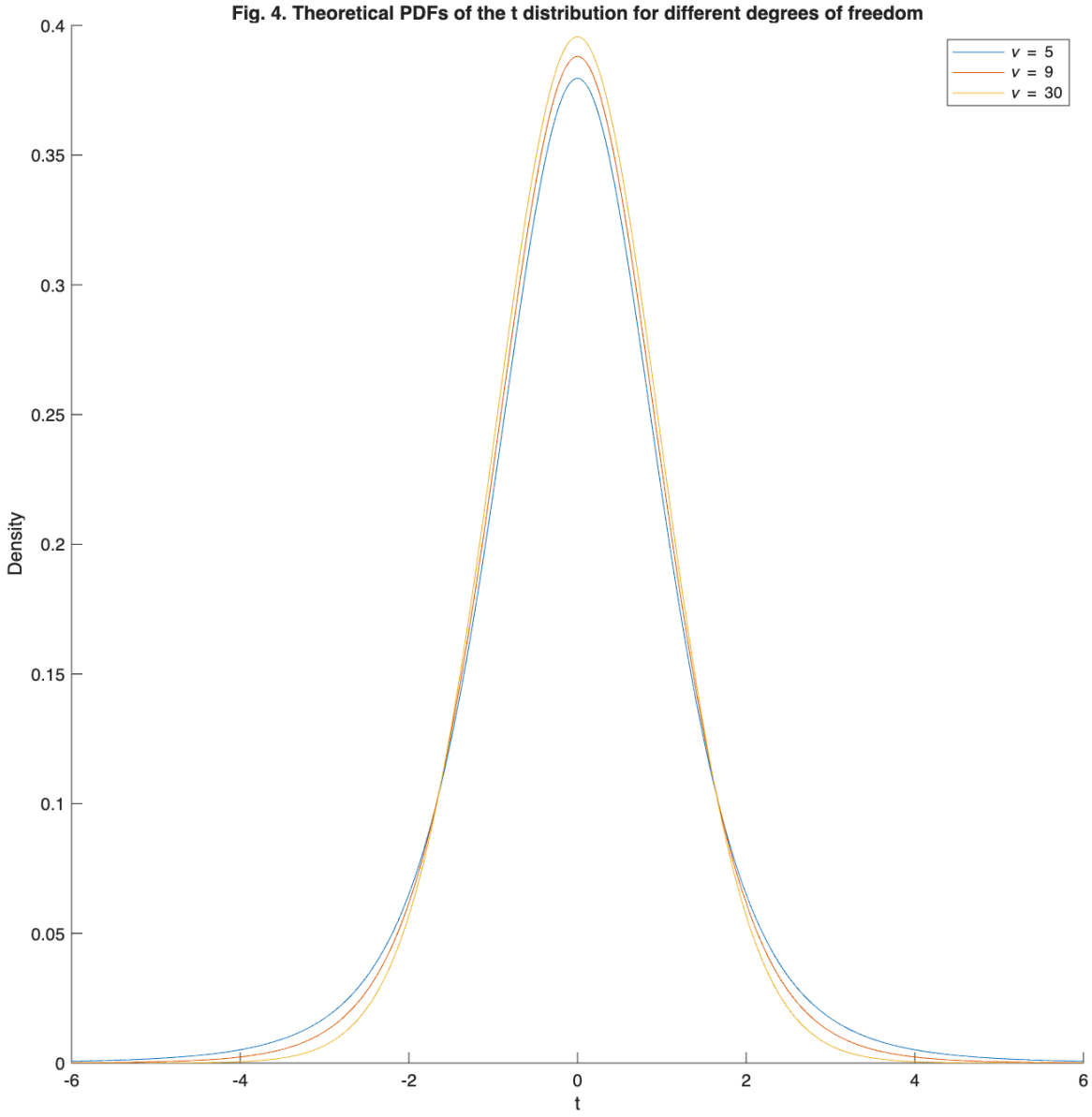


Figure 4: Limiting distributions with different degrees of freedom

8. Final notes

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